

AIR FORCE TECHNICAL APPLICATIONS CENTER (AFTAC)

BROAD AGENCY ANNOUNCEMENT (BAA)

**NEW INITIATIVES FOR AFTAC
SYSTEMS DEVELOPMENT DIRECTORATE**

**25 Apr 2024
AFTAC_BAA_2024**

Table of Contents

1. INTRODUCTION AND BACKGROUND	3
2. PURPOSE.....	3
3. BAA APPROACH AND OVERVIEW	4
4. GENERAL INFORMATION.....	4
5. SUBMISSION INFORMATION	9

1. INTRODUCTION AND BACKGROUND

1.1 The Air Force Technical Applications Center (AFTAC) provides the U.S Department of Defense's (DoD) time dominant scientific and technical information to the right person for decision advantage supporting international nuclear treaty compliance and strategic, operational, and tactical warfighters. Its mission is to detect nuclear explosions anywhere on earth, the atmosphere, or in space. AFTAC monitors for, collects, analyzes, characterizes, and exploits space, missile, atmospheric, subsurface, and hydrological events to identify violations to several nuclear and non-nuclear treaties worldwide. AFTAC provides 24/7 monitoring to support integrated tactical warning and attack assessment (ITW/AA) and Nuclear Force Management (NFM) for the United States command authorities, providing rapid reporting through USSTRATCOM. AFTAC operates the US National Data Center (US NDC), which is the US arm of the International Monitoring System (IMS) supporting the Comprehensive Test Ban Treaty Organization (CTBTO). The US NDC and the IMS support international efforts to limit nuclear testing and proliferation worldwide. AFTAC supports the International Atomic Energy Agency with material sample analysis capabilities, providing accurate measurements to support nonproliferation efforts. Finally, AFTAC supports consequence management and incident response for suspected or confirmed nuclear related incidents, supporting the National Technical Nuclear Forensics (NTNF) mission, and provides program management for Ballistic Missile Technical Collection (BMTC) radar platforms. AFTAC provides national authorities quality technical measurements to monitor nuclear treaty compliance and develops advanced proliferation monitoring technologies to preserve our nation's security. AFTAC also performs research and development of new proliferation detection technologies to enhance or assist treaty verification to limit the proliferation of weapons of mass destruction.

1.2 AFTAC manages research, development, test and evaluation (RDT&E) efforts to advance state-of-the-art technical capabilities in nuclear treaty monitoring, advanced detection technologies to support the counter-proliferation of weapons of mass destruction and other applied technologies.

2. PURPOSE

2.1 This Broad Agency Announcement (BAA), as contemplated in Federal Acquisition Regulation (FAR) 6.102(d)(2) and 35.016, is a competitive solicitation of proposals focused on research and development of technologies to meet AFTAC's mission. AFTAC seeks the best approaches to meet the technology objectives within the capability areas listed below. Detailed topics associated with each capability area appear in Attachment 6.

3. BAA APPROACH AND OVERVIEW

- 3.1 This BAA seeks the best RDT&E approaches to meet various technology objectives of AFTAC. The Government encourages proposals that span a wide spectrum of possible technical and business solutions in response to the specific technology topics stated in Attachment 6. The Government reserves the right to award funding to any combination of proposals which offer the best overall value to the Government, and to oversee all processes and approaches once initiated. Awards of efforts because of this announcement will be in the form of contracts or other transactions (OT) as codified in 10 U.S.C. §2371b (section 2371b). It is anticipated that the awards under this BAA will generally be Cost or Cost-Plus Fixed Fee (CPFF). Cost reimbursement contracts require successful offerors to have an accounting system considered adequate for tracking costs applicable to the contract.
- 3.2 Funding for participation in this program is highly competitive, and the cost of proposed technologies should be considered. Awards resulting from this BAA will be made based on the evaluation results of a phased proposal process described in Section 5. The Government reserves the right to fund all, some, one, or none of the proposals submitted; may elect to fund only part of a submitted proposal; and may incrementally fund any or all awards under this BAA. All awards are subject to the availability of funds.
- 3.3 The acquisition team intends to manage communications in a manner best designed to facilitate the submittal of optimal proposals while not providing any unfair competitive advantages. Technical questions may be addressed up to the white paper submittal. Questions determined to provide a competitive advantage will be answered at www.sam.gov. Dialogue between prospective offerors and Government representatives is encouraged until submission of proposals. Discussions with any of the points of contact shall not constitute a commitment by the Government to subsequently fund or award any proposed effort. Only Contracting Officers are legally authorized to commit the Government. The points of contact associated with this BAA are specified in Section 4.1.

4. GENERAL INFORMATION

- 4.1 POINT OF CONTACTS. Questions regarding the administrative or technical content of this BAA must be addressed to both david.dike.2@us.af.mil and aftac.baa.mailbox@us.af.mil
- 4.2 WITHIN SCOPE MODIFICATIONS. Potential offerors are advised that due to the inherent uncertainty of research and development efforts, awards resulting from this announcement may be modified during performance to make within scope changes, to include but not limited to, modifications which increase overall contract ceiling amount. The Government reserves the right to extend performance to all, some, or none of the selected solutions and decisions will be made based on the availability of

government funds. Only a Contracting Officer can legally bind a contract/agreement modification on behalf of the United States Government.

4.3 NON-GOVERNMENT ADVISORS AND SOURCE SELECTION SENSITIVE INFORMATION. Non-government Advisory and Assistance Services (A&AS) contractors may participate in the evaluation of white paper submissions, when necessary. This will be handled on a case-by-case basis and non-government A&AS contractors shall have signed a non-disclosure agreement (NDA) with the Government. The Government understands that the information provided in response to this BAA is presented in confidence and may contain trade secrets or commercial or financial information, and it agrees to protect such information from unauthorized disclosure to the maximum extent permitted or required by law. AFTAC/SDX will treat all submissions against this BAA as source selection information. During the evaluation process, submissions may be handled by non-government A&AS contractors for technical evaluation(s). All Government support contractors shall be bound by appropriate NDAs.

4.4 SUPPLIER PERFORMANCE RISK SYSTEM (SPRS). To receive a DoD contract every contractor will be required to perform a Basic NIST SP 800-171 DoD Assessment. Offerors should be aware of the provision DFARS 252.204-7019 Notice of NIST SP 800-171 DoD Assessment Requirements and clause DFARS 252.204-7020 NIST SP 800-171 DoD Assessment Requirements. The offeror shall have a current assessment (i.e., not more than 3 years old) for each covered contractor information system that is relevant to the offer, contract, task order, or delivery order. DoD Assessments are described in the NIST SP 800-171 DoD Assessment Methodology located at <https://www.acq.osd.mil/asda/dpc/cp/cyber/safeguarding.html#nistSP800171>. Once the Basic NIST SP 800-171 DoD Assessment is complete, each contractor will need to follow the instructions here <https://www.sprs.csd.disa.mil/> to enter the results into SPRS on the PIEE website.

4.5 DATA RIGHTS IN TECHNICAL DATA, COMPUTER SOFTWARE AND COMPUTER SOFTWARE DOCUMENTATION.

Technical Data: Unlimited Rights

Noncommercial Computer Software (NCS): Unlimited Rights

NCS Documentation: Unlimited Rights

Commercial Computer Software Rights: Customary Commercial License consistent with Federal statutes and regulations.

AFTAC is engaged in the discovery, development, and integration of war-fighting technologies for our air, space, and cyberspace forces. As such, rights in non-commercial technical data and NCS developed or delivered under this contract are of significant concern to the Government. The Government will therefore evaluate any restrictions on the use of non-commercial technical data, NCS, and NCS documentation which could result in transition difficulty or less-than full and open competition for subsequent development of this technology.

In accordance with DFARS 252.227-7013(b)(1) and 252.227-7014(b)(1), the

Government shall receive unlimited rights in all non-commercial technical data and computer software developed exclusively with Government funds.

In accordance with DFARS 252.227-7013(b)(2) and DFARS 252.227-7014(b)(2), the Government shall receive Government Purpose Rights in all non-commercial technical data and computer software developed with mixed funding. “Developed with mixed funding” means, “development was accomplished partially with costs charged to indirect cost pools and/or costs not allocated to a government contract, and partially with costs charged directly to a government contract. Offerors that propose delivery of noncommercial technical data, NCS, or NCS documentation subject to Government Purpose Rights should fully explain how a portion of the data was developed at private expense. Specifically, offerors must explain what noncommercial technical data, NCS, or NCS documentation developed with costs charged to indirect cost pools and/or costs not allocated to a government contract will be incorporated, how the incorporation will benefit the program, and address whether those portions or processes are segregable.

Offerors that propose delivery of noncommercial technical data with Limited Rights, NCS with Restricted Rights, or NCS documentation with Limited Rights will be considered. Proposals should fully explain what noncommercial technical data, NCS, or NCS documentation developed with costs charged to indirect cost pools and/or costs not allocated to a government contract will be incorporated and how the incorporation will benefit the program and whether those portions or processes are segregable.

Offerors shall include the data rights assertions as required by DFARS 252.227-7017, Identification and Assertion of Restrictions on the Government’s Use, Release, or Disclosure of Technical Data or Computer Software. The assertions list is due at the time of proposals. Assertions must be completed with specificity. Each assertion must identify the technical data or computer software to be delivered, and the associated item, component, or process developed exclusively or partially at private expense to which it pertains. Nonconforming data rights assertion lists will not be accepted until submitted in accordance with DFARS 252.227-7017.

Offerors shall mark each page of its white paper that contains proprietary information. It is the Government’s intention to acquire only that necessary to satisfy agency needs and this will be negotiated on a case-by-case basis. An offeror may provide with its white paper assertions to restrict, use, release or disclosure of data and/or computer software and computer software documentation that will be provided during contract performance. Rules governing these assertions are prescribed in DFARS clauses 252.227-7013, 7014, and 7017 and may be accessed at www.acquisition.gov

Terms used in this section are defined in the clauses at 252.227-7013, Rights in Technical Data-Noncommercial Items, and 252.227-7014, Rights in Noncommercial Computer Software and Noncommercial Computer Software Documentation.

THIRD PARTY SOFTWARE (COMMERCIAL AND NONCOMMERCIAL):

If any such software, noncommercial or commercial, is not reasonably identifiable at proposal submission, it must still be approved by the contracting officer prior to incorporation. This obligation to obtain pre-approval from the contracting officer, as described above, continues throughout contract administration.

Noncommercial Computer Software:

DFARS 252.227-7014(d) describes requirements for incorporation of third party noncommercial copyrighted computer software and computer software documentation and is incorporated as follows:

The Contractor shall not, without the written approval of the Contracting Officer, incorporate any copyrighted computer software or computer software documentation in the software or documentation to be delivered under this contract unless the Contractor is the copyright owner or has obtained for the Government the license rights necessary to perfect a license or licenses in the deliverable software or documentation of the appropriate scope set forth in DFARS 252.227-7014(b), and prior to delivery of such—

- (1) Computer software has provided a statement of the license rights obtained in a form acceptable to the Contracting Officer; or
- (2) Computer software documentation has affixed to the transmittal document a statement of the license rights obtained.

In addition, all noncommercial computer software will receive the appropriate level rights set forth in DFARS 252.227-7014(b), which could include: Unlimited rights, GPR, Restricted Rights, or specifically negotiated license.

Commercial Computer Software:

For commercial computer software, the Government will neither accept nor execute a DD Form 250 for such software deliverables until the Contractor obtains from all third-party software suppliers and/or vendors (Licensor) licenses for any commercial computer software to be delivered that are consistent with Federal Statutes, Federal Case Law, and Federal Regulations.

The following is a non-exhaustive list of terms and conditions which are inconsistent with Federal law and shall not be included in the commercial computer software license agreement between the Licensor and the Government:

1. The license shall not subject the Government to a contingent liability or a liability that is indefinite or indeterminate, including but not limited to: indemnification clauses, unilateral price increases, the right to attorney fees, automatic assessment of charges, or automatic renewal provisions. These provisions constitute obligations in advance or in excess of an appropriation and violate the Anti-Deficiency Act.
2. The license shall be governed by Federal Statutes, Federal Case Law, and Federal Regulations, and shall not be subject to the laws or jurisdiction of any municipality, state, or foreign country. The license shall not bind the Government to litigation in a

particular forum or venue or require the Government to participate in arbitration.

3. The license shall not include non-substitution language that would preclude or limit the Government from using another vendor/reseller and/or product to fulfill Government requirements.

4. The Licensors shall not have the authority to unilaterally terminate the license. All remedies available shall be consistent with the Disputes and Termination Clauses in the underlying basic contract.

5. The Licensors shall not have the right to enter the premises or monitor Government networks for the purpose of auditing the use of the license.

6. The Licensors shall not have the authority to control or otherwise influence any litigation between a third party and the Government. The United States Department of Justice has the sole authority to represent the Government in all litigation matters.

7. The Licensors shall not use the fact that the Government is using the Licensors' products in any notification or advertisement to the public (e.g., no publicity rights permitted).

8. The license shall not require automatic updates or give the Licensors the authority to unilaterally replace the software.

Additionally, the Contractor may be required to obtain licenses that comply with the following terms and conditions, based on the Government's needs:

1. [IF APPLICABLE] The license shall not disclaim all warranties through use of an "as is" provision.

2. [IF APPLICABLE] The license shall neither restrict the Government from using the product at various sites nor limit use of the product by various Government agencies or third parties performing work on behalf of the Air Force under the [PROGRAM NAME]. In performance of the [PROGRAM NAME], Government personnel as well as Government contractors may use the software, subject to any negotiated limits on number of users, as applicable.

3. [IF APPLICABLE] [The license shall not limit the Government's use of the software at other Government and Government contractor sites.] [The license shall authorize the Government to use the software at the following sites: [list].

4. [IF APPLICABLE] The license shall not restrict the Government from copying or embedding elements of accessible code into other applications (e.g., nesting code, derivative works).

The Contractor may obtain agreement from the Licensors to insert the clause below in its respective software licenses intended to be transferred to the Government:

"In the event that any of the provisions of the [Software License] are determined to be inconsistent with Federal law or do not otherwise satisfy the Government's needs, the parties to the [Software License] hereby agree that such provisions shall be null and void as they pertain to the Government. Specifically, the following sections are

hereby deleted from the [Software License] [and/or amended as indicated below]:
[Section X: deleted; Section Y: amended as follows [...]]”

If the Licensor will not agree to the terms and conditions cited herein and/or as contained in DFARS 227.72, the Contractor shall retain the current license on behalf of and for the benefit of the US Government if permissible under its license and such use will not subject the Government to the terms of the license. If the software in question is required to be delivered to the Government, the Licensor must grant the Government a sublicense that allows the Government to use the software to meet its requirements.

The Contractor shall provide documentation to clearly correlate or map any commercial computer software to be delivered to:

- a) Contract Line-Item Numbers (CLINS);
- b) Contract Deliverables (CDRLS);
- c) Paragraphs in the statement of work (SOW); and
- d) Portions of any functional block diagrams and/or system architecture diagrams, so that it can be readily determined where certain commercial computer software corresponding to certain software license agreement(s) are physically located on the system to be delivered under the contract.

5. SUBMISSION INFORMATION

5.1 *SUBMISSION OVERVIEW*. Phase I submissions shall include both the Quad Charts and White Papers prepared in accordance with the formats at Attachments 1 and 2 respectively. Phase I submissions can be submitted within 1 year of the date this notice was posted on www.sam.gov and offerors are encouraged to monitor www.sam.gov for any specific calls or amendments to the BAA that may have updated information. AFTAC will invite select Offerors to submit full proposals based on favorable evaluation results of Phase I submissions. Phase II submissions will be requested ONLY by the Contracting Officer and will contain separate submission instructions and timelines within the Request for Proposal (RFP) letter.

5.2 *TWO-PHASED SUBMISSION*. This BAA will be conducted in two phases as follows:

5.2.1 *Phase I*. Interested offerors must submit Quad Charts and White Papers in accordance with instructions provided in this BAA. Phase I submissions will be evaluated against criteria as described in Attachment 7 of this BAA.

5.2.1.1 All Quad Charts should include the information indicated on the sample template in Attachment 1.

5.2.1.2 The White Paper narrative expands on the Quad Chart presentation. Instructions for format and content are in Attachment 2. Each offeror shall include in the white paper a preliminary Technology Readiness Level (TRL) assessment of the proposed technology that conforms to the technology objectives and an expected TRL at the conclusion of the effort. The offeror shall perform a TRL assessment

according to the Department of Defense Technology Readiness Assessment Guidance.

5.2.1.3 White paper and quad chart submissions shall be sent to both david.dike.2@us.af.mil and aftac.baa.mailbox@us.af.mil for review.

5.2.2 *Notification of Selection.* All offerors providing Phase I submissions will be contacted by the Government, either with an email letter informing them that the effort proposed is not of interest to the Government at this time or with a request for a formal cost and technical proposal by a specific date.

5.2.3 *Phase II.* Phase II proposals will only be accepted from Offerors who upon review of Phase I submissions are invited by the Contracting Officer to submit a full proposal. In order to be considered for award under Phase II, offerors must be registered in the System for Award Management (SAM) at www.sam.gov, be considered responsible within the meaning of FAR Part 9.1, Responsible Prospective Contractors, have a satisfactory performance record, and otherwise be eligible for award based on federal law and regulation.

5.2.3.1 *Proposal Submission.* Upon notification of a proposal by the Contracting Officer, offerors have 30 calendar days to prepare and submit a proposal in accordance with the format and instructions at Attachments 3-5. Each proposal shall include a Statement of Work (SOW) and detailed costs. All pages shall be numbered.

5.2.3.2 *Proposal Review, Evaluation and Selection Process.* Each proposal will be evaluated by the Government and the proposal must stand on its own technical merit. Proposals will be evaluated in accordance with the evaluation criteria and process specified in Attachment 7. Pursuant to FAR 35.016(e), the primary basis for selecting proposals for acceptance shall be technical, importance to agency programs, and fund availability. To provide the desired evaluation, qualified Government personnel will conduct reviews and (if necessary) convene panels of experts in the appropriate areas.

5.3 *Rights in Technical Data, Computer Software and Computer Software Documentation.* Offerors shall mark each page of its white paper that contains proprietary information. Refer to paragraph 4.5 for additional information.

ATTACHMENT 1: QUAD CHART FORMAT

The following information must be included in the Phase I Quad Chart. Quad Charts shall not exceed 1-page in landscape view. Quad Charts must follow the formatting guidelines on the slide template below.

BAA FA702224S0001

Project Title:

Date:

Objective: Clear concise (1-2 sentence) description of the goal of the effort ($\geq$ 12-pt font)

Description of Effort: Brief description of the technology proposed for investigations and methods to be used during the course of investigation ($\geq$ 12-pt font).

Benefits: Brief statement that identifies the net advantages of the proposed technology over current practices and competing technologies ($\geq$ 12-pt font)

Challenges:

- Bulleted list of technical or scientific challenges ($\geq$ 12-pt font)

Maturity of Technology: Description of the current maturity of the proposed technology with respect to the Technology Readiness Level ($\geq$ 12-pt font).

Photograph or graphic illustration of concept or technology. Any text must be $\geq$ 12-pt font.

Major Goals and Milestones by Fiscal Year (FY):

- Bulleted list ($\geq$ 12-pt font)
- Deliverables: (e.g., study, written reports, prototype device, or proof of prototype testing)

Proposed Funding (\$K):

Year 1	Year 2	Total
\$000	\$000	\$000

Period of Performance: ($\geq$ 12-pt font).

Contact Info: Name, email, phone ($\geq$ 12-pt font).

ATTACHMENT 2: PHASE I WHITE PAPER FORMAT

The White Paper narrative expands on the Quad Chart presentation and must be 8.5 x 11 inches with one-inch margins, no longer than five (5) pages, single-spaced, with text in Times New Roman, no smaller than 12-point font. All white papers must specify a project title, the Offeror's organization, and the specific topic number associated with the proposal (see Attachment 6). Pages in excess of the page limitation will not be read or evaluated. White Papers must not exceed a file size of 25 MB.

The White Paper must be limited only to further explanation of the information conveyed in the Quad Chart, as deemed necessary by the Offeror. The information should provide sufficient information on the research being proposed (e.g. the hypothesis, theories, concept, approaches, data measurements, and analysis, etc.) to allow for an assessment by a technical expert. Expand as deemed necessary on the following sections of the Quad Chart: Objective; Description of the Effort; Benefits of the Proposed Technology; Technical Challenges; Technology Readiness Level (TRL) of the technology at project start and planned TRL at project completion; Major Goals/Milestones by Government fiscal year. Do not include corporate or personnel qualifications, past experience or any supplemental information not requested in the Quad Chart.

White paper and quad charts shall be sent to david.dike.2@us.af.mil and aftac.baa.mailbox@us.af.mil for review.

ATTACHMENT 3: PHASE II TECHNICAL PROPOSAL FORMAT

The Technical Proposal portion of the Phase II submission shall not exceed 25 pages, excluding cover page and table of contents. Any pages submitted in excess of the 25-page limit will not be considered or evaluated.

Suggested page limitations for individual sections are listed below; however, these are guidelines not strict limits.

- I. ABSTRACT. *[1 page suggested.]*
- II. SCOPE. This proposal is submitted in response to the Air Force Technical Applications Center (AFTAC) Nuclear Treaty Monitoring Broad Agency Announcement (BAA) on.... *[Cite the Project Title. 8 pages suggested.]*
 - a. Objective. *[A clear and concise description of the objective of the proposed project.]*
 - b. Background. *[Provide the necessary technical and scientific background to support the scientific and technical merits of the proposed project.]*
 - c. Programmatics. This effort will support the AFTAC Nuclear Treaty Monitoring initiative *[topic number]* aimed at developing *[capability area or topic title]* technologies or methodologies. *[Describe the Offeror's management plan for the proposed project; list supporting and collaborating organizations and subcontractors; clearly identify roles and responsibilities of each organization and subcontractor supporting the project].*
 - d. Relevance. *[Describe the relevance of the proposed project in terms of AFTAC mission, end-user needs and the technology maturity of the proposed project.]*
- III. CREDENTIALS. *[Describe the Offeror's qualifications to perform the proposed work. Summarize the relevant credentials of the primary performing center and supporting academic and industrial partners. Describe specific examples of similar work performed, and equipment and/or facilities available to perform the proposed work. List summary of the Key Personnel qualifications. Focus on information directly relevant to the proposed work. 4 pages suggested.]*
 - a. Summary of Credentials.
 - b. Summary of Qualifications for Key Personnel.
 - c. Summary of Facilities to Perform the Proposed Work.
- IV. WORK TO BE PERFORMED. *[Provide details of the work to be performed by task and subtask. 9 pages suggested.]*
 - a. General. *[Provide an overview.]*

- b. Detailed Tasks. *[Describe the details of all tasks needed to perform the proposed work. Clearly distinguish base period tasks and option tasks as appropriate.]*

i. BASE PERIOD (YEAR 1)

1. Task 1: Task Title 1 *[Describe what will be accomplished, how the task will be performed, the resources allocated to the task (personnel, hours, materials, etc.), appropriate metrics to measure progress, and deliverables from the task. Describe applicable subtasks. Ensure task description clearly identifies why the task is necessary to accomplish project objectives. Do not include cost details here.]*
2. Task 2: Task Title 2
3. Task N: Task Title N *[Define as many tasks as necessary to achieve the stated objective(s).]*

ii. OPTION 1 (YEAR 2) *[Define as many Option Periods as necessary to achieve the state objective(s). Proposed work shall not exceed 5 years.]*

1. Task N+1: Task Title N+1
2. Task N+2: Task Title N+2
3. Task N+M: Task Title N+M

- V. PERFORMANCE SCHEDULE. *[Depict the tasks and sub-tasks and the duration of performance of each in a Gantt chart or other suitable chart. Briefly describe the proposed period of performance. 2 pages suggested.]*

- VI. REFERENCES. *[List any relevant documents used to develop the technical approach. 1 page suggested]*

ATTACHMENT 4: PHASE II COST PROPOSAL FORMAT

The cost proposal must include cost estimates sufficiently detailed for meaningful evaluation of the cost proposal. The Offeror must include a Cost Breakdown detailing all cost elements, and a Cost Narrative to support the requirements of each cost element. In addition, the Offeror must provide separate cost proposal(s), at the same level of detail as the prime contractor, for each subcontractor or consultant. For pricing purposes, Offerors should assume a contract or agreement start date of approximately ninety (90) days after submission of the proposal.

All proposals are subject to the requirements of the Truth in Negotiations Act (TINA). Any proposal exceeding the threshold listed in Federal Acquisition Regulations (FAR) 15.403-4(a)(1) must submit a certificate of current cost and pricing data in the format described in FAR 15.406-2. The certificate of current cost and pricing data is required with proposal submission, however, after negotiations Offerors may be asked to re-submit an updated certificate.

The responsibility for providing adequate supporting data and attachments lies solely with the Offeror. Further, the Offeror must also bear the burden of proof in establishing reasonableness of proposed costs; therefore, it is in the Offeror's best interest to submit a fully supportable and well-prepared cost proposal. The basis and rationale for all proposed costs should be provided as part of the proposal so that Government personnel can place reliance on the information as current, complete, and accurate.

The Offeror shall prepare the Cost Proposal (except for the narrative information described below) using Microsoft Excel format but shall merge all Cost Proposal documents into one Adobe PDF file for ultimate proposal submission. Include the Topic Number and the Project Title on all pages. The Offeror shall retain the Excel version of all Cost Proposal documents for subsequent submission to the Government should the Offeror's proposal be selected for the award.

- a. Format. Cost proposals shall be formatted as follows.
 - (1) Paper Size – 8.5 X 11-inch paper
 - (2) Margins 1-inch
 - (3) Spacing – single
 - (4) Font – Times New Roman, 12 Point
 - (5) Electronic Software: Microsoft Word, Excel, or Adobe Acrobat
- b. Additional Information: The following information should be contained in a cover sheet attached to the cost proposal:
 - (1) Full company name and address
 - (2) BAA number submitting proposal against
 - (3) Point of contact, name, title, and phone and fax number
 - (4) CAGE code

- (5) UEI number (formerly known as DUNS number)
- (6) Type of contract/agreement proposed (i.e. FFP, CPFF)
- (7) Name, address, and telephone number of the Offeror's cognizant Defense Contract Management Agency (DCMA) administration office.
- (8) Name, address, and telephone number of the Offeror's cognizant Defense Contract Audit Agency (DCAA) audit office.
- (9) Will you require the use of any Government property in the performance of this work? If yes, identify.
- (10) Is this proposal consistent with your established estimating and accounting practices and procedures and FAR Part 31 cost principles? If not, explain.
- c. Cost Element Breakdown: Clear, concise and accurate cost proposals reflect the Offeror's financial plan for accomplishing the effort contained in the technical proposal. All direct costs (e.g., labor, material, travel, and computer) as well as labor and overhead rates should be provided by contractor fiscal year (CFY) unless otherwise specified in the BAA. The cost element breakdown(s) may include the following.
- (1) Direct Labor: Direct labor should be detailed by the number of labor hours, category, and rates in burdened costs.
- (2) Labor and Overhead Rates: Direct labor hours, with their applicable rates, must be broken out by CFY and the bases used clearly identified. The source of labor and overhead rates and all pricing factors should be identified. For instance, if a Forward Pricing Rate Agreement (FPRA) is in existence, that should be noted, along with the Administrative Contracting Officer's (ACO's) name and telephone number. If the rates are based on current experience in your organization, provide the history base used and clearly identify all escalation, by year, applied to derive the proposed rates. If computer usage is determined by a rate, identify the basis used and rationale used to derive the rate.
- (3) Material/Equipment: List all material/equipment items by type and kind with associated costs and advise if the costs are based on vendor quotes, data, and/or engineering estimates; provide copies of vendor quotes and/or catalog pricing data.
- (4) Subcontractor Costs: Offerors must submit all subcontractor proposals and analyses with the cost proposal (See FAR 15.404-3(b)). If the subcontractor will not submit cost and pricing information to the Offeror, this information must be submitted directly to the Government for analysis. On all subcontracts and interdivisional transfers, provide the method of selection used to determine the subcontractor and the proposed contract type of each

subcontract. An explanation shall be provided if the Offeror proposes a different amount than that quoted by the subcontractor. The Offeror's proposal must:

- (a) Identify principal items/services to be subcontracted.
- (b) Identify prospective subcontractors and the basis on which they were selected. If non-competitive, provide selected source justification.
- (c) Identify the type of contractual business arrangement contemplated for the subcontract and provide a rationale for same.
- (d) Identify the basis for the subcontract costs (e.g., firm quote or engineering estimate, etc.).
- (e) Identify the cost or pricing data or information other than cost or pricing data submitted by the subcontractor.
- (f) Provide an analysis of the proposed subcontract in accordance with FAR 15.404-3(b). Provide an analysis concerning the reasonableness, realism and completeness of each subcontractor's proposal. If the analysis is based on comparison with prior prices, identify the basis on which the prior prices were determined to be reasonable. The analysis should include, but not be limited to, an analysis of materials, labor, travel, other direct costs, and proposed profit or fee rates.

(5) Special Tooling or Test Equipment: When special tooling and/or test equipment is proposed, attach a brief description of said items and indicate if they are solely for the performance of this particular contract or project and if they are or are not already available in the Offeror's existing facilities. Indicate quantities, unit prices, whether items are to be purchased or fabricated, whether items are of a severable nature and the basis of the price. These items may be included under direct material in the summary format.

(6) Consultants: When consultants are proposed to be used in the performance of the contract, indicate the specific project or area in which such services are to be used. Identify each consultant, number of hours or days to be used and the consultant's rate per hour or day. State the basis of said rate and give your analysis of the acceptability of the consultant's rate.

(7) Travel: Travel costs must be justified and related to the needs of the project. Identify the number of trips, the destination, and purpose. Travel costs should be broken out by trip with number of travelers, airfare, per diem, lodging, etc.

NOTE: If an Offeror takes exceptions to the requirements called out in the announcement, the exceptions should be clearly stated in the cost proposal.

ATTACHMENT 5: PHASE II STATEMENT OF WORK (SOW) FORMAT

A SOW must be included with the full proposal as part of the Phase II submission. The SOW does not have a page limit but should be approximately 3–5 pages in length and must be a separate and distinct document suitable for incorporation into the procurement instrument. Do not include proprietary data or markings in the SOW. Pages should be numbered and the initial page should have a date (document date) shown under the title.

The proposed SOW must accurately describe the work to be performed. The proposed SOW must also contain a summary description of the technical methodology as well as the task description, but not in so much detail as to make the SOW inflexible. The SOW template should include the following sections and specifically respond to the Statement of Objectives (SOO) the Government provided with the Request for Proposal (RFP):

1. **OBJECTIVE**. This section is intended to give a brief overview of the project and should describe why the work is being pursued and what the Offeror intends to accomplish.
2. **SCOPE**. This section includes a statement of what the SOW covers. This should describe the capability area (Broad Agency Announcement topic) to be investigated, goals, and milestones for the effort.
3. **BACKGROUND**. The Offeror must identify appropriate documents that are applicable to the effort to be performed. This section includes any information, explanations, or constraints that are necessary to understand the project tasks and requirements. This section should describe the effort's relationship to previous and ongoing efforts.
4. **TASKS AND TECHNICAL REQUIREMENTS**. This section contains the detailed description of tasks that are necessary to perform the proposed work. These tasks will represent the work to be performed in any potential contract award, and, thus, shall be considered contractually binding. This section should be developed in an orderly progression and presented in sufficient detail to establish the feasibility of accomplishing the overall program goals. This section should clearly distinguish base period tasks from option period tasks. The sequence of performance must be presented by fiscal year and task order, as in Section IV of the Technical Proposal. The SOW must contain every task to be accomplished. Note that the Government reserves the right to modify and negotiate the language in the SOW prior to any potential contract award.
 - 4.1. Task descriptions must be definite, realistic, and clearly stated. Use the phrase “the contractor shall” whenever the work statement expresses a provision that is binding. Use “should” or “may” whenever it is necessary to express a declaration of purpose. Use “will” in cases where no Offeror requirement is involved (e.g., power will be supplied by the Government. Use active voice in describing work to be performed.
 - 4.2. Do not use acronyms or abbreviations without first defining them. Place

abbreviations and acronyms in parentheses following the first usage of the spelled-out phrase.

- 4.3. If presentations and meetings are identified in the Offeror's schedule, the Offeror must include the following statement in the SOW: "Conduct presentations and meetings at times and places specified in the contract schedule."
- 4.4. The desired project structure consists of base period tasks, focused on demonstrating the feasibility of the technology, and option tasks, which cover design, fabrication, testing, and evaluation of the final product. Execution of optional tasks will be dependent upon and determined by the contractor's performance during and the results produced from the base period tasks.

ATTACHMENT 6: SOLICITATION TOPICS

The AFTAC Systems Development Directorate (AFTAC/SD) is interested in innovative technologies and services related to the areas of mission or technical interest listed below. Submissions should identify innovative solutions that fill capability needs related to the development of AFTAC capabilities for nuclear treaty monitoring, advanced detection technologies to support the counter-proliferation of weapons of mass destruction (WMD), and other applied technologies for both nuclear and non-nuclear event detection. The items, technologies, or services can also directly improve the effectiveness or efficiency of operating and/or maintaining current AFTAC capabilities related to the topics listed below. The level of detail provided for each topic or the order in which they appear is not intended to convey any information regarding relative priority.

There are no specific Technology Readiness Level (TRL) requirements for proposed projects. However, proposed prototypes or solutions should be at a high enough TRL to enable the assessment of the viability, technical feasibility, application, or military utility of the proposed solution and support a rapid transition into operations.

AFTAC intends to fund successful efforts in Fiscal Years 2024 and 2025 (FY24/FY25) with periods of performance in the range of 1-5 years. Exceptions may be considered for individual projects based on technological maturity or operational need. Individual contracts are expected to be funded in the range of \$150,000-\$5,000,000 over the full period of performance, yet these numbers should not be considered strict limits. Proposals aimed at the higher end of this funding range will only be accepted if they demonstrate the ability to deliver capable, innovative solutions that directly impact on the AFTAC mission and offer cost savings or improved efficiencies in current processes or procedures. The Government reserves the right to fund all, some, one or none of the proposals submitted; may elect to fund only part of a submitted proposal; and may incrementally fund any or all awards under this BAA. All awards are subject to the availability of funds.

Topics for each capability area are detailed below.

TOPIC: AFTAC-001, GEOPHYSICS and SEISMOLOGY

AFTAC seeks innovative solutions for analytical techniques in seismology, hydroacoustics, and infrasonic phenomenology. This topic includes:

- Application of the use of lidar, digital imagery and cartographic products, such as geology maps, into event analysis.
- Application of the use of infrasound data for treaty monitoring purposes.
 - Modeling signal propagation through realistic atmospheres
 - Event location estimation with low uncertainty
 - Yield estimation with low uncertainty
 - Event discrimination
 - Evaluation of techniques for wind noise suppression with existing and experimental sensor technologies

- Modeling of seismic, infrasound and hydroacoustic signal-source scaling for a variety of sources or source conditions.
 - Fully contained explosions, underground or underwater
 - Partially contained explosions, underground or underwater
 - Surface explosions, over ground or water
 - Atmospheric explosions, over ground or water
 - Frequency modulated sources or Doppler signals from air or ground vehicles
- Wave propagation and attenuation modeling with Full Waveform Earth Models.
 - Improve the prediction of waveforms and amplitudes at local and regional distances
 - Use of cepstral analysis for event depth, multiple event identification
- Managing ingestion and flow of multiple data sources for increased capabilities.
 - Handle secure ingestion of data from multiple external sources
 - Handle movement of data across a guard to higher level networks
 - Including auditing of data flows
 - Data conversion between data formats
- Application of new innovative sensors, improve qualification methods for new and existing sensors.
- Innovative hardware and software solutions for modern, flexible, and sustainable geophysical data acquisition systems and data processing; this can include (but is not limited to) new technologies to improve workstation processing, video display capabilities, and long-term data storage and retrieval.

TOPIC: AFTAC-002, ATMOSPHERE AND SPACE

AFTAC seeks innovative and novel solutions for the collection and analysis of signal observables from space, atmospheric and surface nuclear detonations (NUDET) that directly leads to operational technology. This topic includes:

- Modeling of any and all phenomenologies and processes between and including the NUDET or surrogate source and the recorded signal produced by the sensor detecting front end.
 - Modeling of the NUDET or surrogate signal generation, including interaction with its environment that results in a transmitted and detectable signal. Any and all phenomenologies are of interest (e.g., optical, RF, neutron, gamma-ray, x-ray).
 - Signal transport from the transmitted signal(s) to the signal arrival at the detecting sensor element. All intervening media are of interest (e.g., atmosphere, clouds, ionosphere, and space environment).
 - Conversion of the detectable signal at the sensor's front end into a data signal that can be processed (e.g., sensing element design, physics and engineering).
- Development and demonstration of computing environments and computational techniques for optimal NUDET signal analysis that are or can be used in an operational environment (e.g., High Performance Computing (HPC) applications, physics-based algorithms, and artificial intelligence (AI) tools to assist operational analysis).
- Translation of analysis algorithms and techniques into executable code.
- Implementation of sustainable, common ground architecture to rapidly integrate new detection platforms.
- Development, demonstration and implementation of technologies that can be used to extend the mission life of on-orbit payloads and their host space vehicles.

TOPIC: AFTAC-003, DIRECTED ENERGY WEAPON (DEW)

AFTAC seeks innovative solutions for emerging DEW technologies in all domains. This topic includes:

- Advanced imaging and sensing, pointing and tracking systems, data collection and analysis, and modeling and simulation of RF and laser technologies.
- Develop methods and tools to enable and improve the analysis of multi-phenomena data sets.
- Enhanced tools and methods for clutter suppression, coherent processing and for extracting low level signals of interest.

TOPIC: AFTAC-004, NUCLEAR AND NUCLEAR-RELATED MATERIALS ANALYSIS

AFTAC seeks innovative solutions to support trace-level to high-activity analyses of environmental samples for radionuclides. This topic includes:

- Operational laboratory and field methods to monitor potential radionuclide releases of “zero yield” events to confirm that nuclear fission did not occur during a foreign weapon test
- Detection of lower concentrations of radionuclides
- Nuclear debris collection

TOPIC: AFTAC-005, MATERIALS MODELING AND ANALYSIS

AFTAC seeks innovative solutions to support advanced meteorological and environmental modeling efforts. This topic includes:

- Advanced meteorological modeling tools and techniques, transport and dispersion modeling techniques, uncertainty estimation, and high-performance computing.
- Meteorological model development, advanced weather data assimilation techniques, ensemble modeling and uncertainty estimation techniques, transport and dispersion modeling, and high-performance computational techniques.
- Advanced simulation and modeling techniques, environmental fate of compounds of interest, the identification of environmental constituents that may have an impact on US personnel or the performance of equipment and issues related to homeland defense.
- Transport and dispersion modeling, computational chemistry, and process engineering.

TOPIC: AFTAC-006, ENTERPRISE-WIDE INTEGRATION AND ARCHITECTURE MODERNIZATION

AFTAC seeks innovative solutions to support the integration of data across disparate monitoring phenomenologies and modernization of hardware/software architectures. This topic includes:

- New solutions to integrate data access and discoverability across varying monitoring phenomenologies to lower detection thresholds and/or increase efficiency of current operations.
- Technologies to modernize hardware/software architectures or implement improved software design and accrediting processes to more flexibly meet mission needs.

TOPIC: AFTAC-007, ENTERPRISE ASSET AND LIFECYCLE MANAGEMENT IMPROVEMENTS

AFTAC seeks innovative solutions that can provide enterprise-wide asset management visibility as well as improve our lifecycle management capabilities.

- Increase accuracy of forecasting of requirements and scheduling of procurements through the use and exploitation of supply chain demand data.
- Supply chain management, specifically: Automated systems to reduce/eliminate inefficiency, improve asset control, decrease touchpoints and minimize inventory.
- Automated identification and reporting of components and systems with substandard reliability.

TOPIC: AFTAC-008, HIGH PERFORMANCE AND SECURE CLOUD COMPUTING AND DEV OPS SOLUTIONS FOR DATA INGESTION, EXPLOITATION, AND DISSEMINATION

- Offer the ability to process data and perform complex calculations at high speeds, addressing the three critical HPC components (Computation, Network, and Storage).
- Utilize parallel processing to perform modeling, simulation and other scientific computing tasks.
- In a secure connection and networking environment, offer cloud computing that delivers the following hosted services over the Internet: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS).
- Software architecture and applications that can be utilized in the continuous development, continuous integration, and continuous delivery of software products that are being developed in an Agile software development environment.
- Machine Learning (ML) and Artificial Intelligence (AI) Solutions.

ATTACHMENT 7: EVALUATION CRITERIA AND SELECTION PROCESS

The Government will review each Quad Chart/White Paper (Phase I) and select the offerors that have the greatest potential to meet the needs of AFTAC technology requirements based on the topic areas stated in Attachment 6. Initially, a determination will be made if each offeror is technically qualified and has a comprehensive understanding to undertake the development of the technology based on the information stated in the Quad Chart/White Paper. The Government will determine the most technically competent and capable of qualified offerors using the criteria below.

- Criteria (Factors) for Selecting Quad Chart/White Paper for Full Proposal Submission:
 - Technical merit/applicability,
 - Importance to agency programs, and
 - Fund availability
- Relative Importance: All Factors are of equal importance
- Method of Evaluation: Peer or scientific review